

2016.0 RANGE ROVER (LG), 100-00

GENERAL INFORMATION

DESCRIPTION AND OPERATION

DRIVER/PASSENGER DOOR MODULE (DDM/PDM)

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable.
Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle

NOTES:

- If the control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the warranty policy and procedures manual, or determine if any prior approval programme is in operation, prior to the installation of a new module/component
- Generic scan tools may not read the codes listed, or may read only 5-digit codes. Match the 5 digits from the scan tool to the first 5 digits of the 7-digit code listed to identify the fault (the last 2 digits give extra information read by the manufacturer-approved diagnostic system)
- When performing voltage or resistance tests, always use a digital multimeter accurate to three decimal places and with a current calibration certificate. When testing resistance, always take the resistance of the digital multimeter leads into account
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
- If diagnostic trouble codes are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always check for loose connections and corroded terminals
- Where an 'on demand self-test' is referred to, this can be accessed via the 'diagnostic trouble code monitor' tab on the manufacturers approved diagnostic system
- Check DDW for open campaigns. Refer to the corresponding bulletins and SSM's which may be valid for the specific customer complaint and carry out the recommendations as required

The table below lists all diagnostic trouble codes (DTCs) that could be logged in the Driver/Passenger Door Module (DDM/PDM). For additional diagnosis and testing information refer to the relevant diagnosis and testing section.

For additional information, refer to: [Driver Door Module](#) (419-10 Multifunction Electronic Modules, Diagnosis and Testing).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
B1087-83	LIN Bus "A" - Value of signal protection calculation incorrect	Door switchpack internal failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new door switchpack
B1087-86	LIN Bus "A" - Signal invalid	Door switchpack internal failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new door switchpack
B1087-87	LIN Bus "A" - Missing message	- Door switchpack power or ground circuit open circuit, high resistance - LIN bus circuit short circuit to ground, short circuit to power, open circuit, high resistance - Door switchpack internal failure	- Refer to the electrical circuit diagrams and check the door switchpack power and ground circuits for open circuit, high resistance - Refer to the electrical circuit diagrams and check the LIN bus circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new door switchpack
B1087-88	LIN Bus "A" - Bus off	LIN bus circuit short circuit to ground, short circuit to power, open circuit, high resistance	Refer to the electrical circuit diagrams and check the LIN bus circuit for short circuit to ground, short circuit to power, open circuit, high resistance

B11C3-07	Power Striker Open Relay - Mechanical failures	Door soft close mechanical failure	Check the mechanical operation of the door soft close components
B11C3-11	Power Striker Open Relay - Circuit short to ground	Closing motor circuit short circuit to ground	Refer to the electrical circuit diagrams and check the closing motor circuit for short circuit to ground
B11C3-12	Power Striker Open Relay - Circuit short to battery	Closing motor circuit short circuit to power	Refer to the electrical circuit diagrams and check the closing motor circuit for short circuit to power
B11C3-13	Power Striker Open Relay - Circuit open	Closing motor circuit open circuit, high resistance	Refer to the electrical circuit diagrams and check the closing motor circuit for open circuit, high resistance
B1207-12	Crash Input Hardwired Signal - Circuit short to battery	Crash event circuit short circuit to power	Refer to the electrical circuit diagrams and check the crash event circuit for short circuit to power
B1207-14	Crash Input Hardwired Signal - Circuit short to ground or open	Crash event circuit short circuit to ground, open circuit, high resistance	Refer to the electrical circuit diagrams and check the crash event circuit for short circuit to ground, open circuit, high resistance
B1207-29	Crash Input Hardwired Signal - Signal invalid	Crash event circuit short circuit to ground, short circuit to power, open circuit, high resistance	Refer to the electrical circuit diagrams and check the crash event circuit for short circuit to ground, short circuit to power, open circuit, high resistance
B12BC-11	Mirror Position Sensor Supply - Circuit short to ground	Door mirror position sensor power circuit short circuit to	Refer to the electrical circuit diagrams and check the door mirror position sensor power circuit for short circuit to ground

		ground	
B12C4-11	Mirror Turn Indicator - Circuit short to ground	Side repeater lamp circuit short circuit to ground	Refer to the electrical circuit diagrams and check the side repeater lamp circuit for short circuit to ground
B12C4-15	Mirror Turn Indicator - Circuit short to battery or open	Side repeater lamp circuit short circuit to power, open circuit, high resistance	Refer to the electrical circuit diagrams and check the side repeater lamp circuit for short circuit to power, open circuit, high resistance
B12D8-11	Mirror Heater Output - Circuit short to ground	Door mirror heater circuit short circuit to ground	Refer to the electrical circuit diagrams and check the door mirror heater circuit for short circuit to ground
B12D8-15	Mirror Heater Output - Circuit short to battery or open	Door mirror heater circuit short circuit to power, open circuit, high resistance	Refer to the electrical circuit diagrams and check the door mirror heater circuit for short circuit to power, open circuit, high resistance
B12DA-11	Blind Spot Warning Indicator - Circuit short to ground	Blindspot monitoring system alert/status LED circuit short circuit to ground	Refer to the electrical circuit diagrams and check the blindspot monitoring system alert/status LED circuit for short circuit to ground
B12DA-15	Blind Spot Warning Indicator - Circuit short to battery or open	Blindspot monitoring system alert/status LED circuit short circuit to power, open circuit, high resistance	Refer to the electrical circuit diagrams and check the blindspot monitoring system alert/status LED circuit for short circuit to power, open circuit, high resistance
B1326-11	External Door Handle LED -	Exterior door handle LED	Refer to the electrical circuit diagrams and check the exterior

	Circuit short to ground	circuit short circuit to ground	door handle LED circuit for short circuit to ground
B1326-15	External Door Handle LED - Circuit short to battery or open	Exterior door handle LED circuit short circuit to power, open circuit, high resistance	Refer to the electrical circuit diagrams and check the exterior door handle LED circuit for short circuit to power, open circuit, high resistance
B1378-12	Mirror Horizontal Position Sensor - Circuit short to battery	Door mirror position sensor X axis signal circuit short circuit to power	Refer to the electrical circuit diagrams and check the door mirror position sensor X axis signal circuit for short circuit to power
B1378-14	Mirror Horizontal Position Sensor - Circuit short to ground or open	Door mirror position sensor X axis signal circuit short circuit to ground, open circuit, high resistance	Refer to the electrical circuit diagrams and check the door mirror position sensor X axis signal circuit for short circuit to ground, open circuit, high resistance
B137C-12	Mirror Vertical Position Sensor - Circuit short to battery	Door mirror position sensor Y axis signal circuit short circuit to power	Refer to the electrical circuit diagrams and check the door mirror position sensor Y axis signal circuit for short circuit to power
B137C-14	Mirror Vertical Position Sensor - Circuit short to ground or open	Door mirror position sensor Y axis signal circuit short circuit to ground, open circuit, high resistance	Refer to the electrical circuit diagrams and check the door mirror position sensor Y axis signal circuit for short circuit to ground, open circuit, high resistance
B147F-07	Power Window Motor - Mechanical failures	Window movement obstructed	Check the open/close operation of the window. Check the installation of the window seal. Calibrate the window and re-test. If the fault

			persists, install a new window seal and re-test. If the fault persists, install a new window regulator
B147F-72	Power Window Motor - Actuator stuck open	▪ Door window regulator motor circuit short circuit to ground, short circuit to power ▪ Door module internal failure	▪ Refer to the electrical circuit diagrams and check the door window regulator motor circuit for short circuit to ground, short circuit to power ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new door module
B147F-73	Power Window Motor - Actuator stuck closed	▪ Door window regulator motor circuit short circuit to ground, short circuit to power ▪ Door module internal failure	▪ Refer to the electrical circuit diagrams and check the door window regulator motor circuit for short circuit to ground, short circuit to power ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new door module
B147F-92	Power Window Motor - Performance or incorrect operation	▪ Window movement obstructed	▪ Check the open/close operation of the window. Check the installation of the window seal. Calibrate the window and re-test. If the fault persists, install a new window seal and re-test. If the fault persists, install a new door window regulator
B147F-97	Power Window Motor - Component or system operation obstructed or blocked	▪ Door window regulator motor circuit open circuit, high resistance ▪ Door window regulator motor position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check the door window regulator motor circuit for open circuit, high resistance ▪ Refer to the electrical circuit diagrams and check the door window regulator motor position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance

B1480-29	Window Position Sensor - Signal invalid	▪ Door window regulator motor circuit open circuit, high resistance ▪ Door window regulator motor position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check the door window regulator motor circuit for open circuit, high resistance ▪ Refer to the electrical circuit diagrams and check the door window regulator motor position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
B1481-11	Window Blind Motor - Circuit short to ground	▪ Door blind motor circuit short circuit to ground	▪ Refer to the electrical circuit diagrams and check the door blind motor circuit for short circuit to ground
B1481-12	Window Blind Motor - Circuit short to battery	▪ Door blind motor circuit short circuit to power	▪ Refer to the electrical circuit diagrams and check the door blind motor circuit for short circuit to power
B1481-13	Window Blind Motor - Circuit open	▪ Door blind motor circuit open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check the door blind motor circuit for open circuit, high resistance
B1481-92	Window Blind Motor - Performance or incorrect operation	▪ Door blind obstructed ▪ Door blind not calibrated	▪ Remove any obstructions from the door blind. Calibrate the door blind ▪ Calibrate the door blind
B1482-11	Door Double Locking Motor - Circuit short to ground	▪ Double locking motor circuit short circuit to ground	▪ Refer to the electrical circuit diagrams and check the double locking motor circuit for short circuit to ground
B1482-15	Door Double Locking Motor - Circuit short to	▪ Double locking motor circuit short	▪ Refer to the electrical circuit diagrams and check the double locking motor circuit for short

	battery or open	circuit to power, open circuit, high resistance	circuit to power, open circuit, high resistance
B1483-11	Door Central Locking Motor - Circuit short to ground	Central door locking motor circuit short circuit to ground	Refer to the electrical circuit diagrams and check the central door locking motor circuit for short circuit to ground
B1483-15	Door Central Locking Motor - Circuit short to battery or open	Central door locking motor circuit short circuit to power, open circuit, high resistance	Refer to the electrical circuit diagrams and check the central door locking motor circuit for short circuit to power, open circuit, high resistance
B1484-11	Lock Status LED - Circuit short to ground	Security system status indicator circuit short circuit to ground	Refer to the electrical circuit diagrams and check the security system status indicator circuit for short circuit to ground
B1484-15	Lock Status LED - Circuit short to battery or open	Security system status indicator circuit short circuit to power, open circuit, high resistance	Refer to the electrical circuit diagrams and check the security system status indicator circuit for short circuit to power, open circuit, high resistance
B1488-11	Up/Down Mirror Motor - Circuit short to ground	Door mirror vertical adjustment actuator circuit short circuit to ground	Refer to the electrical circuit diagrams and check the door mirror vertical adjustment actuator circuit for short circuit to ground
B1488-15	Up/Down Mirror Motor - Circuit short to battery or open	Door mirror vertical adjustment actuator circuit short circuit to power, open	Refer to the electrical circuit diagrams and check the door mirror vertical adjustment actuator circuit for short circuit to power, open circuit, high resistance

		circuit, high resistance	
B1489-11	Left/Right Mirror Motor - Circuit short to ground	Door mirror horizontal adjustment actuator circuit short circuit to ground	Refer to the electrical circuit diagrams and check the door mirror horizontal adjustment actuator circuit for short circuit to ground
B1489-15	Left/Right Mirror Motor - Circuit short to battery or open	Door mirror horizontal adjustment actuator circuit short circuit to power, open circuit, high resistance	Refer to the electrical circuit diagrams and check the door mirror horizontal adjustment actuator circuit for short circuit to power, open circuit, high resistance
B148A-11	Deployable Door Handle Motor - Circuit short to ground	Deployable door handle actuator circuit short circuit to ground	Refer to the electrical circuit diagrams and check the deployable door handle actuator circuit for short circuit to ground
B148A-12	Deployable Door Handle Motor - Circuit short to battery	Deployable door handle actuator circuit short circuit to power	Refer to the electrical circuit diagrams and check the deployable door handle actuator circuit for short circuit to power
B148A-13	Deployable Door Handle Motor - Circuit open	Deployable door handle actuator circuit open circuit, high resistance	Refer to the electrical circuit diagrams and check the deployable door handle actuator circuit for open circuit, high resistance
B148B-11	Folding Mirror Motor - Circuit short to ground	Door mirror powerfold actuator circuit short circuit to ground	Refer to the electrical circuit diagrams and check the door mirror powerfold actuator circuit for short circuit to ground
B148B-15	Folding Mirror Motor - Circuit short to battery or	Door mirror powerfold actuator circuit	Refer to the electrical circuit diagrams and check the door mirror powerfold actuator circuit

	open	short circuit to power, open circuit, high resistance	for short circuit to power, open circuit, high resistance
B148C-11	Approach Lamp - Circuit short to ground	Approach lamp circuit short circuit to ground	Refer to the electrical circuit diagrams and check the approach lamp circuit for short circuit to ground
B148D-11	Memory Status LED - Circuit short to ground	Seat memory switchpack status indicator circuit short circuit to ground	Refer to the electrical circuit diagrams and check the seat memory switchpack status indicator circuit for short circuit to ground
B148D-15	Memory Status LED - Circuit short to battery or open	Seat memory switchpack status indicator circuit short circuit to power, open circuit, high resistance	Refer to the electrical circuit diagrams and check the seat memory switchpack status indicator circuit for short circuit to power, open circuit, high resistance
P0604-42	Internal Control Module Random Access Memory (RAM) Error - General memory failure	Door module internal failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new door module
P0605-42	Internal Control Module Read Only Memory (ROM) Error - General memory failure	Door module internal failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new door module
P1603-44	EEPROM Malfunction - Data memory failure	Door module internal failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new door module
U0010-88	Medium Speed CAN	Medium speed CAN	Using the manufacturer approved diagnostic system, perform a CAN

	Communication Bus - Bus off	bus (body) circuit short circuit to ground, short circuit to power, open circuit, high resistance	network integrity test. Refer to the electrical circuit diagrams and check the medium speed CAN bus (body) circuit for short circuit to ground, short circuit to power, open circuit, high resistance
U0011-81	Medium Speed CAN Communication Bus Performance - Invalid serial data received	Invalid data received from another control module via the medium speed CAN bus (body)	Using the manufacturer approved diagnostic system, check the snapshot data to determine the invalid data source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0011-82	Medium Speed CAN Communication Bus Performance - Alive/sequence counter incorrect/not updated	Invalid data received from another control module via the medium speed CAN bus (body)	Using the manufacturer approved diagnostic system, check the snapshot data to determine the invalid data source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0011-87	Medium Speed CAN Communication Bus Performance - Missing message	Missing message from another control module via the medium speed CAN bus (body)	Using the manufacturer approved diagnostic system, check the snapshot data to determine the missing message source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0300-00	Internal Control Module Software Incompatibility - No sub type information	Door module is not configured correctly	Using the manufacturer approved diagnostic system, re-configure the door module with the latest level software
U200D-11	Control Module Output Power A - Circuit short to ground	Door switchpack circuit short circuit to ground	Refer to the electrical circuit diagrams and check the door switchpack power circuit for short circuit to ground
U200E-	Control Module	Approach	Refer to the electrical circuit

11	Output Power B - Circuit short to ground	lamp circuit short circuit to ground	diagrams and check the approach lamp power circuit for short circuit to ground
U2010-11	Switch Illumination - Circuit short to ground	Door switchpack illumination circuit short circuit to ground	Refer to the electrical circuit diagrams and check the door switchpack illumination circuit for short circuit to ground
U2010-15	Switch Illumination - Circuit short to battery or open	Door switchpack illumination circuit short circuit to power, open circuit, high resistance	Refer to the electrical circuit diagrams and check the door switchpack illumination circuit for short circuit to power, open circuit, high resistance
U201A-56	Control Module Main Calibration Data - Invalid/incomplete configuration	Car configuration file mismatch with vehicle specification	NOTE: After updating the car configuration file, set the ignition to on and wait 30 seconds before clearing the DTCs Using the manufacturer approved diagnostic system, check and update the car configuration file as necessary. Clear the DTCs and retest